

OAK vs. Unconstrained Additive GPs: Cross Terms, Marginal Likelihood Simplification, and Low-Variance Expansions

Contents

1	Introduction	2
2	Review of classic Additive Kernels	2
2.1	Additive Gaussian Process Kernels	2
2.2	The Identifiability Problem and Spurious Expansions	2
3	Reviewing Orthogonal Additive Kernels (OAK)	3
3.1	Definition and Orthogonality Constraint	3
3.2	Constrained Kernels	3
4	Covariance Structure: Cross Terms and Diagonalisation	3
4.1	Unconstrained Additive Kernel Covariance	3
4.2	OAK Covariance and Block-Diagonality in Function Space	4
5	Marginal Likelihood Simplification	4
5.1	Marginal Likelihood in GP Models	4
5.2	Unconstrained Versus OAK Marginal Likelihood	4
5.3	Analytical Consequence: Reduced Redundancy	5
6	Low-Variance Functions and Generalization	5
6.1	Concept of Low Variance in Function Space	5
6.2	How OAK Promotes Low-Variance Representations	5
7	Summary	6

1 Introduction

In Lu’s Revisited paper, it was unclear how the OAK’s *property* of unique decompositions led to the subset consisting of lower-order terms whilst Additive did not. As we know that the space of functions from Additive should be a superset of the functions from OAK, we wish to investigate the reasons behind the split of the space and also any subsequent properties.

In unconstrained additive GPs (e.g., Duvenaud et al., 2011), the lack of identifiability often leads to *spurious expansions* where the same effect is redundantly represented in multiple interaction terms. This document presents a detailed analysis of how the Orthogonal Additive Kernel (OAK) framework, by imposing orthogonality constraints on the partial functions, removes these redundant expansions, simplifies the covariance structure, and yields a simpler marginal likelihood landscape that promotes low-variance, well-regularized models. We believe that this simpler marginal likelihood may explain the subset split occurring mentioned previously.

2 Review of classic Additive Kernels

2.1 Additive Gaussian Process Kernels

An additive kernel decomposes the overall kernel into a sum over functions on subsets of the input dimensions. For a d -dimensional input $x = (x_1, \dots, x_d)$, a simple additive kernel takes the form:

$$k_{\text{add}}(x, x') = \sum_{i=1}^d k_i(x_i, x'_i),$$

and more generally, one may include interaction terms:

$$k_{\text{add}}(x, x') = \sum_{u \subseteq \{1, \dots, d\}} \sigma_u^2 \prod_{i \in u} k_i(x_i, x'_i),$$

where σ_u^2 controls the contribution of the subset u .

2.2 The Identifiability Problem and Spurious Expansions

In unconstrained additive kernels, there is no unique way to decompose a function. For example, given a true function

$$f(x_1, x_2) = f_{\{1\}}(x_1) + f_{\{2\}}(x_2),$$

an unconstrained additive GP can represent it equivalently as:

$$f(x_1, x_2) = [f_{\{1\}}(x_1) + \Delta] + [f_{\{2\}}(x_2) - \Delta],$$

or by absorbing parts of the main effects into the interaction term. This non-uniqueness leads to *spurious expansions* where redundant representations inflate the effective complexity.

3 Reviewing Orthogonal Additive Kernels (OAK)

3.1 Definition and Orthogonality Constraint

The OAK framework restricts the additive function space by enforcing orthogonality among the partial functions. For every subset $u \subseteq \{1, \dots, d\}$, the corresponding function $f_u(x_u)$ is constrained to have zero mean with respect to the input measure:

$$\int f_u(x_u) p(x_u) dx_u = 0.$$

This constraint forces a unique (FANOVA) decomposition of $f(x)$ as:

$$f(x) = \sum_{u \subseteq \{1, \dots, d\}} f_u(x_u),$$

with

$$\langle f_u, f_v \rangle_{L^2(p)} = 0 \quad \text{for } u \neq v.$$

3.2 Constrained Kernels

For a base kernel $k_i(x_i, x'_i)$ defined on feature x_i , the constrained (orthogonal) kernel is defined as:

$$\tilde{k}_i(x_i, x'_i) = k_i(x_i, x'_i) - \frac{E_p[k_i(x, x_i)] E_p[k_i(x, x'_i)]}{E_{p,p}[k_i(x, x')]},$$

which guarantees that functions drawn from $\tilde{k}_i$ satisfy:

$$\int f_i(x_i) p(x_i) dx_i = 0.$$

For a subset u , the multi-dimensional constrained kernel is obtained as the product:

$$\tilde{k}_u(x_u, x'_u) = \prod_{i \in u} \tilde{k}_i(x_i, x'_i).$$

4 Covariance Structure: Cross Terms and Diagonalisation

4.1 Unconstrained Additive Kernel Covariance

In the unconstrained additive GP, the overall covariance is given by:

$$K_{\text{add}}(x, x') = \sum_u \sigma_u^2 \prod_{i \in u} k_i(x_i, x'_i).$$

Because there are no constraints, the partial functions f_u and f_v (with $u \neq v$) may exhibit nonzero cross-covariance:

$$\text{Cov}[f_u(x), f_v(x')] \neq 0.$$

This nonzero cross-covariance leads to entangled contributions in the prior and complicates the hyperparameter landscape.

4.2 OAK Covariance and Block-Diagonality in Function Space

In OAK, by contrast, the orthogonality constraints imply that

$$\mathbb{E}[f_u(x)f_v(x')] = 0 \quad \text{for } u \neq v.$$

Thus, the GP prior factorizes into independent blocks for each subset u . While the full covariance matrix K_{OAK} over data points is generally not diagonal (since correlations within each partial function persist), the decomposition is *block-diagonal* in the sense that there is no cross-correlation between different additive components. Mathematically, one can write:

$$K_{\text{OAK}}(x, x') = \sum_u \tilde{k}_u(x_u, x'_u),$$

with

$$\text{Cov}[f_u(x), f_v(x')] = 0 \quad \text{if } u \neq v.$$

This clean separation of the function space reduces the risk of redundant representation.

5 Marginal Likelihood Simplification

5.1 Marginal Likelihood in GP Models

The marginal likelihood in a GP model is defined as:

$$p(y | X) = \int p(y | f, X) p(f | X) df,$$

where $p(f | X)$ is the GP prior and $p(y | f)$ is the likelihood (often Gaussian).

5.2 Unconstrained Versus OAK Marginal Likelihood

In the unconstrained additive GP, the prior is over a large, redundant function space. The effective kernel,

$$K_{\text{add}} = \sum_u \sigma_u^2 \prod_{i \in u} k_i(x_i, x'_i),$$

includes off-diagonal (cross-covariance) terms arising from the non-unique decompositions. These additional degrees of freedom lead to a more complex marginal likelihood surface with multiple equivalent local optima.

For OAK, the prior is restricted to functions satisfying the orthogonality constraints. The corresponding kernel,

$$K_{\text{OAK}} = \sum_u \sigma_u^2 \prod_{i \in u} \tilde{k}_i(x_i, x'_i),$$

has the property that the additive components are independent. This yields a marginal likelihood of the form:

$$p(y | X, \theta)_{\text{OAK}} = \frac{1}{(2\pi)^{n/2} |K_{\text{OAK}}|^{1/2}} \exp \left(-\frac{1}{2} y^\top K_{\text{OAK}}^{-1} y \right).$$

Because the components do not “double count” the same effect, the hyperparameter landscape is simpler and the marginal likelihood is more peaked around the best decomposition. In other words, OAK effectively regularizes the model by excluding spurious, redundant expansions.

5.3 Analytical Consequence: Reduced Redundancy

Suppose in an unconstrained model there exist two decompositions that yield the same overall function:

$$f(x) = f_1(x_1) + f_2(x_2) = [f_1(x_1) + \Delta] + [f_2(x_2) - \Delta].$$

Because of this redundancy, the prior integrates over an unnecessarily large function space. In OAK, the constraint

$$\int f_i(x_i)p(x_i)dx_i = 0$$

forces $\Delta = 0$ and uniquely identifies f_1 and f_2 . This elimination of degenerate directions in the prior reduces the effective complexity and improves the marginal likelihood’s behavior.

6 Low-Variance Functions and Generalization

6.1 Concept of Low Variance in Function Space

A key intuition is that *low-variance functions* are less likely to overfit and are more stable. In the GP framework, the total (pointwise) variance is given by the trace of the covariance matrix:

$$\text{Var}(f) = \text{tr}(K).$$

(Must double check this from GPML page 18 *Joint Predictions* sections)

In unconstrained additive GPs, the possibility of redundant expansions can inflate this trace by duplicating the same effect in multiple components.

6.2 How OAK Promotes Low-Variance Representations

By enforcing orthogonality, OAK ensures that each component contributes uniquely. This prevents a situation in which the same pattern is “counted twice”, thus keeping the effective variance allocated to each component lower. In turn, this simplification regularizes the model and leads to better generalization. More formally, let $f(x) = \sum_u f_u(x_u)$ with the orthogonality property:

$$\text{Var}(f(x)) = \sum_u \text{Var}(f_u(x_u)).$$

In the unconstrained case, cross-covariance terms would contribute additional variance:

$$\text{Var}(f(x)) = \sum_u \text{Var}(f_u(x_u)) + \sum_{u \neq v} \text{Cov}(f_u(x_u), f_v(x_v)).$$

Since OAK forces $\text{Cov}(f_u, f_v) = 0$ for $u \neq v$, it avoids this “variance inflation,” which is advantageous for model simplicity and generalization.

7 Summary

In summary, the key differences between the unconstrained additive GP kernel and OAK are:

- (1) **Elimination of Redundant Expansions:** OAK’s orthogonality constraint uniquely decomposes the function, preventing the shuffling of constant or similar components among different partial functions.
- (2) **Simpler Covariance Structure:** Although the full covariance matrix over data points is not literally diagonal, in function space the additive components become independent (block-diagonal), leading to no cross-covariance among distinct feature subsets.
- (3) **Marginal Likelihood Simplification:** The restricted prior in OAK yields a simpler marginal likelihood surface by avoiding redundant parameterizations. This results in a more peaked and well-behaved hyperparameter landscape.
- (4) **Promotion of Low-Variance Functions:** By disallowing double counting, OAK effectively regularizes the model, ensuring that the total variance is not inflated by spurious interactions. This helps in better generalization.

These mathematical and conceptual insights explain why OAK often leads to a better optimization outcome in terms of marginal likelihood and improved generalization performance compared to unconstrained additive GP kernels.